

3. REQUIREMENTS ENGINEERING

3.1 Requirements Elicitation

To understand how the university clinic actually operates day to day, the team relied on a structured interview as the primary method of gathering requirements. Before any interview could take place, the group's instructor first sent a formal request letter to the University Clinic to secure permission, explaining that the activity was for the academic requirements of MediSync and that any information shared would be used responsibly and kept confidential where needed. Once approval was granted, the team prepared a data gathering questionnaire divided into six phases, Clinic Background and General Information, Patient Arrival and Queuing, Patient Records and Documentation, Consultation and Prescriptions, Inventory and Analytics, and a final wrap up phase requesting sample copies of the clinic's existing forms. Organizing the questions this way let the interview move naturally through the clinic's entire workflow instead of jumping between unrelated topics.

The interview itself was conducted with Nurse Kyle Guntang of the university clinic. He was chosen as the respondent because, as one of the five nurses on duty, he has direct, firsthand knowledge of nearly every process the system needs to support, from patient intake and triage to medicine inventory and monthly reporting. The team was present as a group throughout the session, asking questions and taking notes together.

A live interview was chosen over a written survey mainly because the clinic's processes involve a lot of conditional detail that depends on the situation. How a case is handled, for instance, differs depending on whether it involves a minor, an urgent condition, or a routine clearance, and this kind of branching information is much easier to draw out through real time follow up questions than through a fixed set of survey items. Being present also let the team ask immediate clarifying questions whenever an answer opened up a new detail worth exploring, such as how occupancy limits are enforced or how a medical certificate is worded when a student's symptoms do not fully justify one.

The information gathered from the interview directly shaped the project's requirements. Details about the clinic's reliance on paper folders and the recurring problem of misplaced or duplicated records pointed to the need for an electronic health record module. The description of how students currently wait without any visible queue, relying only on a Microsoft Form that gives no sense of position or wait time, established the need for a live queuing and appointment system. The manual, end of day counting of medicine stock and the handwritten inventory logbooks likewise confirmed the need for an automated inventory and analytics dashboard, while the process for issuing medical certificates and prescriptions informed the requirements for the system's documentation module. Nurse Kyle also agreed to share blank copies of the clinic's current forms, which the team is using as a reference to make sure the digital versions built into MediSync match the fields the staff already rely on.

3.2 User Requirements

The following requirements describe what each type of user needs MediSync to help them accomplish, grouped by user type.

Students need to check the clinic's doctor and dentist schedules online and book visits in advance to avoid wasted trips and long waiting lines, and they need to input their initial symptoms to help with triage sorting and track their live queue number so they know when their turn is coming up.

Employees and outside visitors within the campus vicinity need a straightforward way to be recorded when they need care even without an existing student account, whether by requesting a visit ahead of time or registering on the spot when they arrive, and they need the same ability as students to check their own queue status once they are in the system.

Clinic nurses need to record patient vital signs directly into a digital profile to eliminate misplaced paper folders, and they need to view real time medicine stock counts and organize the waiting line based on medical urgency.

The university physician needs to view a patient's past health history and create digital prescriptions that deduct supplies from clinic stock automatically, and needs to generate official medical certificates directly from consultation notes to eliminate handwritten slips.

The university dentist needs the same kind of digital access suited to dental care, viewing a patient's dental history and logging treatment notes directly instead of relying on separate paper charts, and needs to manage the dental appointment schedule so students and staff can see when dental visits are available.

The system administrator needs to manage user accounts and assign the proper access level to each type of user, so confidential clinic records stay protected and every person only sees the parts of the system their role actually requires.

3.3 Functional Requirements

This section lists the specific functions MediSync must perform, drawn from the clinic interview and the modules established in Chapter 2. Table 3.1 presents each requirement with a unique identifier and a description of what the system should do.

Table 3.1. Functional Requirements

ID	Functional Requirement	Description
FR-01	The users, including students, employees, nurses, dentists, and doctors, should be able to log in to their own account.	The system should let each user log in using their own credentials before they can access the functions relevant to their role.
FR-02	The students and employees should be able to view the doctor and dentist schedule.	The system should display the available schedule of doctors and dentists so students and employees can check it before going to the clinic.
FR-03	The students and employees should be able to book an appointment against an available schedule.	The system should let a user reserve an open time slot and prevent the same slot from being booked by more than one person.
FR-04	The clinic staff should be able to register an outside visitor who has no existing account.	The system should let clinic staff create a temporary record for a visitor using the basic details provided at the time of the visit.
FR-05	The clinic staff should be able to confirm a patient's arrival by matching a tap in to an existing appointment.	The system should read a tapped ID and either link it to the matching appointment or add the patient to the walk in queue if no appointment exists.
FR-06	The clinic staff should be able to review a suggested symptom priority level and adjust the queue order.	The system should propose a priority level from the reported symptoms and let clinic staff override it so the more urgent cases can be attended to first.
FR-07	The students and employees should see their queue number for high volume batch processing during peak periods.	The system should show each patient their queue number so they know their turn, especially when there are a lot of patients at once.
FR-08	The clinic staff should be able to store and display the patient's medical record.	The system should keep each patient's medical record on file and let clinic staff view it whenever needed.
FR-09	The clinic staff should be able to track medicine information and its count, and receive an alert when stock is low.	The system should keep track of medicine details and stock levels, and notify clinic staff once the supply is running low.
FR-10	The clinic staff should be able to browse the automatically generated medical certificate based on the patient's consultation record.	The system should generate a medical certificate automatically from the patient's consultation record, which clinic staff can then browse and review before releasing it.

FR-11	The clinic staff should be able to see the dashboard and analytics of patient volume, common illnesses, and inventory.	The system should provide a dashboard showing analytics such as patient volume, common illnesses, and inventory status.
FR-12	The users, including students, employees, dentists, doctors, and nurses, should have different levels of access to the system.	The system should give each user type a different level of access based on their role.
FR-13	The students, employees, and visitors should agree to the terms and conditions before their information is collected.	The system should present a consent notice at registration or first login and require agreement before continuing.
FR-14	The students and employees should receive notifications within the application and through their university Outlook email.	The system should send appointment confirmations, reminders, and clinic announcements through both channels so users do not miss an update.
FR-15	The clinic staff should be able to send a direct message to a student or a parent regarding an appointment or a concern.	The system should let clinic staff compose and send a message tied to a specific patient or case, and keep a record of it.
FR-16	The students should be able to access the health education map and quiz feature.	The system should let a student explore the campus map and answer short health quizzes for their own reference.

3.4 Non-Functional Requirements

This section describes the quality attributes and performance expectations MediSync must meet, covering areas such as performance, security, usability, reliability, compatibility, and availability. Table 3.2 presents each non functional requirement with a unique identifier and a description of the expectation it addresses.

Table 3.2. Non-Functional Requirements

ID	Non-Functional Requirement	Description
NFR-01	The system should respond quickly during normal clinic use.	MediSync should load pages and return queue, record, and inventory information without noticeable delay so that nurses and doctors are not slowed down while attending to students, especially during peak periods such as freshman enrollment.
NFR-02	The system should keep patient information secure and limited by role.	Only users with the correct role, such as a nurse, doctor, dentist, or administrator, should be able to open or edit a student's Electronic Health Record, consistent with the Role Based Access and Security Module, so that private health data is not seen by users who are not authorized to view it.
NFR-03	The system should be easy to use for people with different levels of computer skill.	Since MediSync will be used by students, clinic staff, and student volunteers who are not equally comfortable with technology, the interface should use plain labels, short forms, and a consistent layout so that users can book appointments, record vital signs, or check the queue without needing extensive training.
NFR-04	The system should stay reliable throughout the clinic's operating hours.	MediSync should remain available and function correctly for as long as the clinic is open, including during busy periods with a high volume of bookings and queue updates, so that clinic operations are not interrupted by system errors or slowdowns.
NFR-05	The system should log users out automatically after a period of inactivity.	Because clinic computers may be shared among staff, MediSync should end an inactive session after a set period so that patient records and other sensitive information are not left open and accessible to the next person who uses the computer.
NFR-06	The system should work properly within the university's local network.	Since MediSync depends on the university's local area network rather than the public internet, the system should function consistently for all users connected within the campus network, and any slowdown or interruption should be traceable to the network rather than the system itself.

NFR-07	The system should keep the Medicine Inventory Tracker and Digital Queue information accurate and up to date.	Whenever a medicine is dispensed or a patient's queue status changes, the system should reflect the update immediately, so that clinic staff are always working with the current stock count and current queue order instead of outdated information.
NFR-08	The system should give clinic staff clear and understandable error messages.	If a user enters incorrect information or a required field is missing, such as when recording vital signs or updating stock, the system should show a simple message explaining what needs to be corrected, so that staff can fix the issue quickly without confusion.
NFR-09	The system should handle patient data in line with data privacy regulations.	Since MediSync stores health related information covered by the Data Privacy Act of 2012, access to a record should always require a valid role and, for new users, a completed consent step, and any information shared with school leaders through analytics should stay limited to aggregate numbers rather than individual identities.

3.5 Specific Requirements

This section provides a detailed description of the major modules and functions of MediSync. The requirements are based on the clinic interview, the current clinic workflow, the identified problems with the paper based process, and the proposed features of the system.

3.5.1 User Authentication and Role Management

MediSync shall provide secure login functionality for students, employees, and authorized clinic personnel. Each user shall have their own account credentials before accessing the system. The system shall provide different permissions based on the user's role. Clinic personnel may include nurses, doctors, dentists, and other authorized clinic staff. Each role shall only have access to the functions and patient information required for their responsibilities.

Students and employees shall only be able to access their own account and permitted personal clinic information, and shall not be able to access, modify, or view the records belonging to other users. Outside visitors within the campus vicinity who do not have an existing account shall instead be registered manually by authorized clinic personnel at the time of their visit, using the basic details the visitor provides rather than a self service login.

Before a student, employee, or visitor's information is collected for the first time, the system shall present a terms and conditions and consent notice explaining how their data will be used, and shall require their agreement before proceeding with registration or booking.

The system shall automatically log out inactive users after a specified period of inactivity to protect patient information when clinic computers are shared.

3.5.2 Electronic Health Records

MediSync shall replace the clinic's physical patient folders with centralized electronic health records. The clinic currently retrieves patient folders manually from filing cabinets, and misplaced folders can result in duplicate records.

Authorized clinic personnel shall be able to create, view, and update patient records through the system. Each student's record shall be associated with a unique student account or identifier to help prevent duplicate records. The system shall store the student's consultation history and other approved medical information in one centralized record.

Students shall be allowed to view selected information from their own records, including medical history, prescriptions, medical certificates, and appointment history. Students shall not be permitted to modify medical records unless a specific function is later approved by the clinic.

3.5.3 Appointment and Clinic Schedule Management

MediSync shall provide an appointment management module that allows students to view available schedules and request appointments. Students shall be able to view the availability of the clinic's doctor and dentist before selecting an appointment. This addresses the current problem in which schedules are announced through Facebook and some students still visit the clinic when the doctor is unavailable.

The system shall display available appointment schedules based on the schedules entered by authorized clinic personnel. Clinic personnel shall be able to create, modify, and remove available appointment schedules. The system shall prevent students from booking unavailable or already occupied appointment slots. The system shall record the appointment status, such as scheduled, completed, cancelled, or other statuses defined by the clinic.

3.5.4 Priority Queue Management

MediSync shall provide a digital priority queue for patients who use the clinic. The current clinic workflow follows triage, where urgent cases are prioritized over non urgent cases. Examples identified during the interview include accidents, bleeding, and fainting, while routine concerns such as coughs, red eyes, and standard check ups can wait.

The system shall use three priority levels. High priority refers to a non critical case identified as requiring attention sooner than medium and low priority cases. Medium priority refers to a case that requires clinic attention but can wait after high priority patients. Low priority refers to a routine or less urgent case that can reasonably wait for its turn.

The system shall use the patient's reported symptoms to suggest an initial priority level through its triage classification, which clinic personnel shall review and may adjust based on the patient's actual condition and the clinic's triage guidelines. The queue shall arrange patients according to the confirmed priority level and the queue rules defined by the clinic.

When a patient with a confirmed appointment arrives and taps their ID, the system shall match that tap to the existing appointment and mark the patient as arrived rather than creating a separate walk in entry, while a patient who arrives without a prior appointment shall be added to the queue directly. Students shall be able to view their queue number and current queue status through their account.

MediSync shall not be used to schedule or manage serious physical injuries or other critical emergencies requiring immediate medical attention. Patients with such conditions shall be instructed to seek urgent in person medical assistance rather than relying on an appointment or digital queue. The system's priority queue is intended to organize clinic cases and assist staff with patient flow, and

it shall not replace professional medical judgment.

3.5.5 Consultation and Vital Signs

MediSync shall provide digital consultation forms based on the actual forms currently used by the clinic. The clinic's existing workflow includes identifying the patient's purpose for visiting, recording vital signs, providing the patient's record to the doctor, and performing the appropriate post consultation intervention.

The system shall allow authorized clinic personnel to enter the required information from the clinic's existing forms directly into MediSync, and to record the patient's vital signs and other required consultation information. Doctors and dentists shall be able to review the patient's relevant information before and during consultation. The completed consultation shall be stored and linked to the student's electronic health record.

3.5.6 Prescription and Medicine Dispensing

MediSync shall provide a prescription and medicine dispensing function for authorized clinic personnel. Authorized personnel shall be able to record prescription information as part of the patient's consultation record. When medicine is dispensed, the system shall record the medicine and corresponding quantity based on the clinic's required documentation.

The medicine dispensing transaction shall be associated with the appropriate patient and consultation. Medicine dispensing information shall also update the inventory information so that the recorded stock quantity remains current.

3.5.7 Medical Certificate Generation

MediSync shall provide a digital medical certificate function using information from the student's consultation record. The clinic currently prepares medical certificates manually, and the proposed system is intended to standardize the process using digitized consultation forms.

The system shall generate the medical certificate using the information recorded during the consultation. Authorized clinic personnel shall be able to review the generated certificate before releasing it to the student. The certificate shall contain only information supported by the consultation record and the fields approved by the clinic. Students shall be able to view their released medical certificates through their account.

3.5.8 Medicine and Supply Inventory Management

MediSync shall provide an inventory module for medicines and medical supplies. The clinic currently records incoming medicines and supplies in physical logbooks and manually counts remaining medicine at the end of each day before entering the information into Excel.

Authorized clinic personnel shall be able to record newly received medicines and supplies. The system shall allow authorized personnel to update inventory quantities when medicine is dispensed, received, adjusted, or otherwise changed according to the clinic's inventory process. The system shall display the current recorded stock level of medicines and supplies, and shall provide a low stock alert when an item's quantity reaches or falls below its configured minimum stock level. Inventory information shall be available for use in the clinic dashboard and monthly reports.

3.5.9 Communication and Notification Management

MediSync shall provide notifications for selected appointment related and clinic related activities, delivered both within the application and through the university's Outlook email, in addition to SMS for the events the clinic specifically approves.

For the initial version, notifications may be triggered by events such as successful appointment booking, appointment confirmation, appointment reminder, appointment cancellation or schedule changes, and general clinic announcements. The specific events and message contents shall be defined and approved by the clinic before implementation.

The system shall send SMS notifications using the student's or employee's registered mobile number, and shall send email notifications using the address already associated with their university account. The system shall not send unnecessary or unauthorized patient information through any notification channel.

MediSync shall also allow clinic personnel to send a direct message to a student or, for a minor student, to a parent or guardian, regarding an appointment, a follow up, or another clinic related concern, and shall keep a record of that communication for reference.

3.5.10 Dashboard and Analytics

MediSync shall provide an automated dashboard for authorized clinic personnel. The clinic currently relies on Excel to produce monthly reports involving consultations, patient demographics, and medicine consumption. The clinic specifically identified an automated dashboard as a desired feature.

The dashboard shall summarize relevant clinic information, including total patient or consultation volume, patient demographics, common illnesses or recorded health concerns, released medical clearances or certificates, medicine consumption, and current inventory status. The system shall use information already recorded in MediSync to generate reports instead of requiring nurses to manually transfer information into a separate Excel file. The system shall allow authorized clinic personnel to generate monthly reports from stored clinic data.

3.5.11 Health Engagement

MediSync shall provide a supplementary health engagement feature for students, presented as a simple explorable campus map paired with short health related quizzes.

This feature is meant to give students an easy and voluntary way to learn basic health information outside of an actual clinic visit, and is not a substitute for a consultation, a diagnosis, or professional medical advice from clinic staff. The system shall track a student's completed quizzes for their own reference, and shall not use engagement with this feature to make any clinical decision about the student.

3.5.12 Data Validation and Error Handling

MediSync shall validate information entered by users before storing it in the database. Required fields shall be identified, and the system shall prevent incomplete submissions when the missing information is required. The system shall display clear error messages when users enter invalid, incomplete, or inconsistent information. Error messages shall explain what needs to be corrected so users can resolve common input errors without unnecessary difficulty.

3.5.13 Centralized Database and Information Management

MediSync shall use a centralized database to store student information, electronic health records, appointments, queue information, consultation records, prescriptions, medical certificates, inventory information, and user accounts. The system shall connect related records so that information entered in one module can be correctly associated with the relevant student and transaction.

For example, a completed consultation shall be connected to the student's medical history, any prescription issued during that consultation, and any medical certificate generated from the consultation. The system shall maintain updated information across its modules so that authorized clinic personnel do not need to maintain separate paper and spreadsheet records for the same transaction. This centralized approach directly addresses the clinic's identified problems with misplaced records, duplicate files, lack of live queue information, and manually prepared inventory and monthly reports.
